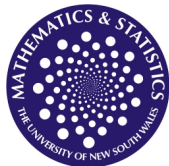




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Lecture Week 3: More priors, Multivariate models, Monte Carlo methods

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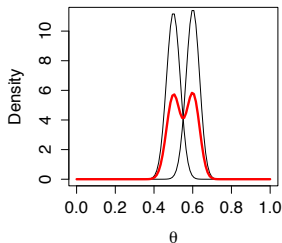
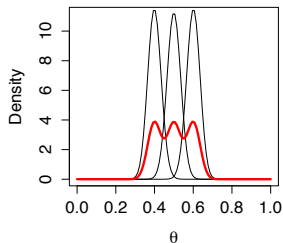
Outline

1. Mixture priors, prior predictive checking.
2. Multivariate models
3. Monte Carlo methods
 - Monte Carlo integration, rejection sampling.

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Mixtures of Priors



Conjugate priors:

- ▶ Mathematically convenient
- ▶ May not be sufficiently flexible to capture prior beliefs

Construct a prior for probability of H when spinning a coin on a table.

- ▶ Tossing coins give good chance of 0.5 prob
- ▶ When spinning, **coin edge imperfections** cause preference for H or T
- ▶ $\Rightarrow$ reasonable prior may favour prob ≈ 0.4 or 0.6
- ▶ May also include prob = 0.5.
- ▶ i.e. **Prior belief may be bi-modal or tri-modal**

Model is: $X|\theta \sim \text{Binomial}(n, \theta)$ so conjugate prior is Beta.
But **Beta does not permit this kind of prior.**

Mixtures of Priors

One solution is to use mixtures of conjugate distributions.

Suppose $\pi_1(\theta), \dots, \pi_k(\theta)$ are all conjugate priors for θ , leading to posterior distributions $\pi_1(\theta|x), \dots, \pi_k(\theta|x)$.

Consider the family of mixture distributions:

$$\pi(\theta) = \sum_{i=1}^k w_i \pi_i(\theta) \quad \sum_{i=1}^k w_i = 1$$

as a prior distribution. Then

$$\begin{aligned} \pi(\theta|x) &= \pi(\theta)L(x|\theta)/m(x) \\ &= \sum_{i=1}^k w_i \pi_i(\theta)L(x|\theta)/m(x) \\ &= \sum_{i=1}^k w_i \pi_i(\theta|x) m_i(x)/m(x) \\ &= \sum_{i=1}^k w_i^* \pi_i(\theta|x) \end{aligned}$$

where $w_i^* = w_i m_i(x)/m(x)$, $m(x) = \sum_{i=1}^k w_i m_i(x)$, $\sum_{i=1}^k w_i^* = 1$. Hence the posterior is in the same conjugate mixture family.

Allows more flexibility for conjugate priors.

Note: We will link back to this result when comparing models in later classes.

Prior predictive checking

You have specified a prior $\pi(\theta)$. Is there a way to check that this represents your prior beliefs about the data?

Posterior predictive distribution:

$$\pi(y|x) = \int \pi(y|\theta)\pi(\theta|x)d\theta$$

= distribution of future data y given observed data x .

Prior predictive distribution:

$$\pi(y) = \int \pi(y|\theta)\pi(\theta)d\theta$$

= distribution of future data y .

Prior predictive procedure

- ▶ Generate samples from $\pi(y)$ given your choice of $\pi(\theta)$.
- ▶ Check if $\pi(y)$ represents your beliefs about the possible data you might observe under the model.
- ▶ If not, modify $\pi(\theta)$ and repeat.

Prior predictive checking

Example: Suppose $\theta \sim \text{Beta}(\alpha, \beta)$ is the prob of an individual having a rare genetic disorder. $X \sim \text{Binomial}(n, \theta)$ is the number of people (out of n independently chosen individuals) that have the disorder.

Prior predictive distribution

$$\pi(y) \propto \int \text{Bernoulli}(\theta) \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta, \quad y = 0, 1.$$

So (algebraically)

$$Pr(y = 0) \propto \int \theta \times \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta = B(\alpha + 1, \beta)$$

$$Pr(y = 1) \propto \int (1-\theta) \times \theta^{\alpha-1} (1-\theta)^{\beta-1} d\theta = B(\alpha, \beta + 1)$$

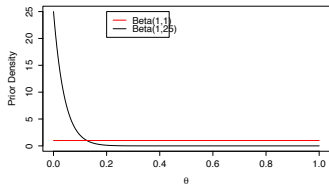
where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$.

$$Pr(y = 0) = B(\alpha + 1, \beta) / [B(\alpha + 1, \beta) + B(\alpha, \beta + 1)] = 1 - Pr(y = 1).$$

Prior predictive checking

Example (cont.):

	True		Monte Carlo	
	$\Pr(y=0)$	$\Pr(y=1)$	$\Pr(y=0)$	$\Pr(y=1)$
Beta(1,1)	0.5	0.5	0.496	0.504
Beta(1,25)	0.038	0.962	0.039	0.961



- ▶ Beta(1,1) prior gives 50-50 chance to have the disorder a priori.
- ▶ This may not represent prior beliefs for a rare disorder.
- ▶ Modify Beta prior parameters until prior probabilities are more realistic.
- ▶ Do not use data from the upcoming analysis to inform this decision!

Outline

1. Mixture priors, prior predictive checking.
2. Multivariate models
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Multivariate models



All examples so far have been 1 parameter e.g., binomial, normal with known variance.

However, virtually every practical problem in statistics involves more than one unknown or unobservable quantity (parameter)

This is where Bayesian methods are **much more straightforward** than their classical counterpart

No new theory is required beyond what we've already seen

Multivariate Models

We have d -dimensional parameter vector $\theta = (\theta_1, \dots, \theta_d)$.

Specify a multivariate prior distribution, $\pi(\theta)$, and compute the posterior in the same manner:

$$\pi(\theta|x) \propto \frac{L(x|\theta)\pi(\theta)}{\int L(x|\theta)\pi(\theta)d\theta}.$$

If we are interested in only 1 or a few parameters, we can obtain the relevant marginal distribution from the joint posterior.

E.g. the marginal distribution of $\theta_1|x$ is obtained by integrating out the other components of θ

$$\pi(\theta_1|x) = \int_{\Theta_d} \dots \int_{\Theta_2} \pi(\theta|x) d\theta_2 \dots d\theta_d.$$

Can do this either a) Algebraically, or b) via Monte Carlo simulation

Q: Why are they called marginal distributions?

Multivariate Models

Bayesian inference for multivariate models requires overcoming the following challenges:

1. Prior specification:

- Priors are now multivariate distributions.
- Involve complex parameter dependencies.
- More complicated to elicit.

2. Computation:

- Required integrals are now more complicated.
- Needs more sophisticated algorithms in absence of conjugates.
- e.g. inversion sampling won't work.

3. Interpretation:

- Structure of the posterior is more complicated.
- Posterior inference and interpretation is more challenging.

Multivariate Models

Example:

- ▶ A machine-produced item is either:
satisfactory ($x = 1$) or unsatisfactory ($x = 2$).
- ▶ The probability of the item being satisfactory depends on
 - a) room temperature ($\theta_1 = 0$: cool, $\theta_1 = 1$: hot) and
 - b) humidity ($\theta_2 = 0$: dry, $\theta_2 = 1$: humid).
- ▶ The probabilities of $x = 1$ under these conditions are given as:

$\Pr(x = 1 \theta_1, \theta_2)$	$\theta_1 = 0$	$\theta_1 = 1$
$\theta_2 = 0$	0.6	0.8
$\theta_2 = 1$	0.7	0.6

Conditional probabilities of item being satisfactory

- ▶ The joint prior distribution of (θ_1, θ_2) is given as:

$\Pr(\theta_1, \theta_2)$	$\theta_1 = 0$	$\theta_1 = 1$
$\theta_2 = 0$	0.3	0.2
$\theta_2 = 1$	0.2	0.3

Prior probabilities of room conditions

Multivariate Models

The **joint posterior distribution** can be calculated as:

		$\theta_1 = 0$	$\theta_1 = 1$	
$Pr(x = 1 \theta_1, \theta_2) \times Pr(\theta_1, \theta_2)$	$\theta_2 = 0$	0.18	0.16	0.34
	$\theta_2 = 1$	0.14	0.18	0.32
		0.32	0.34	0.66

And normalising all probabilities gives:

		$\theta_1 = 0$	$\theta_1 = 1$	
$Pr(\theta_1, \theta_2 x = 1)$	$\theta_2 = 0$	18/66	16/66	34/66
	$\theta_2 = 1$	14/66	18/66	32/66
		32/66	34/66	1

From which we can obtain the marginal distributions:

$$Pr(\theta_1 = 0|x = 1) = 32/66, \quad Pr(\theta_1 = 1|x = 1) = 34/66$$

and

$$Pr(\theta_2 = 0|x = 1) = 34/66, \quad Pr(\theta_2 = 1|x = 1) = 32/66.$$

Multivariate Models

Example: Suppose that $Y_1 \sim \text{Poisson}(\alpha\beta)$ and $Y_2 \sim \text{Poisson}((1 - \alpha)\beta)$ with Y_1 and Y_2 independent given α and β .

Now, suppose our prior information for α and β can be expressed as $\alpha \sim \text{Beta}(p, q)$ and $\beta \sim \text{Gamma}(p + q, 1)$ with α and β independent, for specified hyperparameters p and q .

Then we have the following likelihood

$$\pi(y_1, y_2 | \alpha, \beta) = \frac{\exp(-\alpha\beta)(\alpha\beta)^{y_1}}{y_1!} \times \frac{\exp(-(1 - \alpha)\beta)[(1 - \alpha)\beta]^{y_2}}{y_2!}$$

and the joint prior is

$$\pi(\alpha, \beta) = \frac{\Gamma(p + q)}{\Gamma(p)\Gamma(q)} \alpha^{p-1} (1 - \alpha)^{q-1} \times \frac{e^{-\beta} \beta^{p+q-1}}{\Gamma(p + q)}$$

Multivariate Models

Hence by Bayes' theorem:

$$\begin{aligned}\pi(\alpha, \beta | y_1, y_2) &\propto e^{-\beta} \beta^{y_1+y_2} \alpha^{y_1} (1-\alpha)^{y_2} \alpha^{p-1} (1-\alpha)^{q-1} e^{-\beta} \beta^{p+q-1} \\ &= \beta^{y_1+y_2+p+q-1} e^{-2\beta} \times \alpha^{y_1+p-1} (1-\alpha)^{y_2+q-1}\end{aligned}$$

The marginal posterior distributions are given by

$$\pi(\alpha | y_1, y_2) = \int_0^\infty \pi(\alpha, \beta | y_1, y_2) d\beta \propto \alpha^{y_1+p-1} (1-\alpha)^{y_2+q-1},$$

and

$$\pi(\beta | y_1, y_2) = \int_0^1 \pi(\alpha, \beta | y_1, y_2) d\alpha \propto \beta^{y_1+y_2+p+q-1} e^{-2\beta}$$

so

$$\alpha | y_1, y_2 \sim \text{Beta}(y_1 + p, y_2 + q)$$

and

$$\beta | y_1, y_2 \sim \text{Gamma}(y_1 + y_2 + p + q, 2)$$

Multivariate Methods

Example:

Let $Y_1, \dots, Y_n$ be iid $N(\mu, \sigma^2)$, mean μ and variance σ^2 are unknown.

Take independent Jeffreys priors ($\pi_J(\mu) \propto 1$, $\pi_J(\sigma^2) \propto 1/\sigma^2$):

$$\pi(\mu, \sigma^2) = \pi_J(\mu)\pi_J(\sigma^2) \propto 1/\sigma^2$$

This is a conventional improper prior for this model. Hence

$$\begin{aligned}\pi(\mu, \sigma^2 | y) &\propto \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right) \\ &= \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2\right]\right) \\ &= \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2]\right)\end{aligned}$$

where

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

is the sample variance of the y_i 's.

Multivariate Methods

For the conditional distribution $\mu|\sigma^2, y$ we have

$$\begin{aligned}\pi(\mu|\sigma^2, y) &\propto \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y} - \mu)^2]\right) \\ &\propto \exp\left(-\frac{1}{2\sigma^2}n(\bar{y} - \mu)^2\right) \\ \mu|\sigma^2, y &\sim N(\bar{y}, \sigma^2/n).\end{aligned}$$

For the marginal posterior distribution of σ^2

$$\begin{aligned}\pi(\sigma^2|y) &\propto \int \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2}[(n-1)s^2 + n(\bar{y} - \mu)^2]\right) d\mu \\ &= \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2}(n-1)s^2\right) \int \exp\left(-\frac{1}{2\sigma^2}n(\bar{y} - \mu)^2\right) d\mu \\ &= \sigma^{-(n+2)} \exp\left(-\frac{1}{2\sigma^2}(n-1)s^2\right) \sqrt{2\pi\sigma^2/n} \\ &\propto (\sigma^2)^{-(n+1)/2} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)\end{aligned}$$

Multivariate Methods

More standard (see next slide):

Scale-inverse χ^2 distribution:

$$\pi(x|\nu, \tau^2) = \frac{(\tau^2 \nu / 2)^{\nu/2} \exp\left(-\frac{\nu \tau^2}{2x}\right)}{\Gamma(\nu/2) x^{1+\nu/2}}$$

with $\nu > 0$, $\tau^2 > 0$.

$$\pi(\sigma^2|y) \propto (\sigma^2)^{-(n+1)/2} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)$$

which is a scaled inverse- χ^2 density

$$\sigma^2|y \sim \text{Inv-}\chi^2(n-1, s^2).$$

Multivariate Methods

More helpful:

Inverse gamma distribution – $\text{InvGamma}(\alpha, \beta)$:

$$\pi(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} (1/x)^{\alpha+1} \exp(-\beta/x)$$

with $\alpha > 0$, $\beta > 0$.

If $x \sim \text{Gamma}$ then $z = 1/x \sim \text{InvGamma}$ (tutorial question).

$$\pi(\sigma^2|y) \propto (\sigma^2)^{-(n+1)/2} \exp\left(-\frac{(n-1)s^2}{2\sigma^2}\right)$$

which is an inverse gamma density

$$\sigma^2|y \sim \text{InvGamma}\left(\frac{n-1}{2}, \frac{(n-1)s^2}{2}\right).$$

Multivariate Methods

The marginal posterior of $\mu|y$ is

$$\pi(\mu|y) \propto \int \sigma^{-(n+2)} \exp \left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2] \right) d\sigma^2$$

Substitute $z = A/(2\sigma^2)$ where $A = (n-1)s^2 + n(\bar{y} - \mu)^2$.

$$dz = -\frac{A}{2\sigma^4} d\sigma^2, \quad \sigma^2 = A/(2z).$$

Hence

$$\begin{aligned} \pi(\mu|y) &\propto \int \sigma^{-(n+2)} \exp \left(-\frac{1}{2\sigma^2} [(n-1)s^2 + n(\bar{y} - \mu)^2] \right) d\sigma^2 \\ &= \int -(A/2z)^{-(n+2)/2} \exp(-z) \frac{2(A^2/(2z)^2)}{A} dz \\ &\propto A^{-n/2} \int z^{(n-2)/2} \exp(-z) dz \\ &\propto ((n-1)s^2 + n(\bar{y} - \mu)^2)^{-n/2} \\ &\propto \left(1 + \frac{n(\bar{y} - \mu)^2}{(n-1)s^2} \right)^{-n/2} \end{aligned}$$

Multivariate Methods

$$\pi(\mu|y) \propto \left(1 + \frac{n(\bar{y} - \mu)^2}{(n-1)s^2}\right)^{-n/2}$$

t distribution:

$$\pi(x|\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$$

with $\nu > 0$.

Hence

$$\left. \frac{\mu - \bar{y}}{s/\sqrt{n}} \right| y \sim t_{n-1}.$$

Multivariate Methods

Comment #1: Sampling from the joint posterior distribution is easy, because of the closed form factorisation:

$$\pi(\mu, \sigma^2 | y) = \pi(\mu | \sigma^2, y) \pi(\sigma^2 | y) \\ \text{Normal} \times \text{Inv-}\chi^2 \text{ (or Normal} \times \text{InvGamma)}.$$

Hence, first draw $\sigma^2 | y$ (marginal) and then $\mu | y, \sigma$ (conditional).

Comment #2: A full conjugate analysis for this model is also possible:

$$\mu | \sigma^2 \sim N(\mu_0, \sigma^2 / \kappa_0) \\ \sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2) \text{ (or } \sim \text{InvGamma}(\alpha_0, \beta_0)).$$

This gives the joint prior density

$$\pi(\mu, \sigma^2) \propto \sigma^{-1} (\sigma^2)^{-(\nu_0/2+1)} \exp\left(-\frac{1}{2\sigma^2} [\nu_0 \sigma_0^2 + \kappa_0 (\mu_0 - \mu)^2]\right) \\ (\mu, \sigma^2) \sim \text{Normal-Inv-}\chi^2(\mu_0, \sigma_0^2 / \kappa_0, \nu_0, \sigma_0^2),$$

or

$$\pi(\mu, \sigma^2) \propto \sigma^{-1} (\sigma^2)^{-(\alpha_0+1)} \exp\left(-\frac{1}{2\sigma^2} [2\beta_0 + \kappa_0 (\mu_0 - \mu)^2]\right) \\ (\mu, \sigma^2) \sim \text{Normal-InvGamma}(\mu_0, \sigma_0^2, \alpha_0, \beta_0).$$

Multivariate Methods

Example: Multivariate regression (Tutorial question)

Suppose $x_1, \dots, x_n \in \mathbb{R}^d$ and x_i has a Multivariate Gaussian distribution $N_d(x|\mu, \Sigma)$. Then its probability density function is

$$\pi(x|\mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

where $\mu \in \mathbb{R}^d$ is a mean vector, and Σ is a $d \times d$ positive semi-definite (covariance) matrix.

The precision matrix $\Lambda = \Sigma^{-1}$ ($\tau = 1/\sigma^2$) is also positive semi-definite.

Univariate		Multivariate	
τ	Conjugate = Gamma	Λ	Conjugate = Wishart
σ^2	Conjugate = InvGamma	Σ	Conjugate = InvWishart

Tutorial question ↑

Multivariate Methods

Wishart distribution

The density function of a $W_d(\Lambda|\nu, A)$ is

$$\pi(\Lambda|\nu, A) = \frac{|A|^\nu}{\Gamma_d(\nu)} |\Lambda|^{\nu-(d+1)/2} \exp[-\text{tr}(A\Lambda)],$$

- ▶ Degrees of freedom $\nu \in (\frac{d-1}{2}, \infty)$ (real)
- ▶ Scale parameter $A \sim R^{d \times d}$ (a positive semi-definite matrix).
- ▶ When $d = 1$, this is just the Gamma distribution.

Inverse Wishart distribution

The density function of a $IW_d(\Sigma|\nu, A)$

$$\pi(\Sigma|\nu, A) = \frac{|A|^\nu}{\Gamma_d(\nu)} |\Sigma|^{-\nu-(d+1)/2} \exp(-\text{tr}(A\Sigma^{-1})).$$

- ▶ When $d = 1$, this reduces to the InvGamma distribution.

Multivariate Methods

Example: Multivariate regression

Recall:

$$x_i \sim N(\theta, \sigma^2), \theta \sim N(b, d^2), \tau = 1/\sigma^2, c = 1/d^2$$

$$\theta|x \sim N\left(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau}\right).$$

$$x_i \sim N_d(\mu, \Sigma):$$

The conjugate prior for μ when Σ is known, is $N_d(\mu_0, \Sigma_0)$.

The resulting posterior for $\mu|x$ is $N_d(\hat{\mu}, \hat{\Sigma})$, where

$$\hat{\mu} = (\Sigma_0^{-1} + n\Sigma^{-1})^{-1}(\Sigma_0^{-1}\mu_0 + n\Sigma^{-1}\bar{x})$$

and

$$\hat{\Sigma} = (\Sigma_0^{-1} + n\Sigma^{-1})^{-1}.$$

Clear analogy with univariate result.

Multivariate Methods

Example: Multivariate regression

The conjugate prior for Σ when μ is known, is $IW_d(\Sigma|\nu, A)$.

The resulting posterior for Σ is $IW_d(\Sigma|\hat{\nu}, \hat{A})$, where

$$\hat{\nu} = \nu + \frac{n}{2}$$

and

$$\hat{A} = A + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^\top. \quad (\text{matrix})$$

Again, a clear analogy with univariate result.

Outline

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Recap: Monte Carlo Integration

When working with posterior distributions,
almost all inference can be expressed as an integral.

This is most obvious for expectations:

$$\mathbb{E}_f[X] = \int xf(x)dx.$$

So given a sample $x^{(1)}, \dots, x^{(N)} \sim f(x)$, we can approximate the theoretical mean by the sample mean to give

$$\mathbb{E}_f[X] \approx \frac{1}{N} \sum_{i=1}^N x^{(i)}.$$

Recap: Monte Carlo Integration

Generalising this argument (Law of Large Numbers):

Suppose we wish to calculate

$$\mathbb{E}_f[g(X)] = \int g(x)f(x)dx$$

with respect to f .

Then given a sample $x^{(1)}, \dots, x^{(N)} \sim f(x)$, we can approximate the expectation via

$$\mathbb{E}[g(X)] \approx \frac{1}{N} \sum_{i=1}^N g(x^{(i)})$$

which is an unbiased estimate.

Monte Carlo Integration

Example: Approximate

$$\int_0^1 \exp(x^2) dx.$$

Solution:

The integral can be re-expressed as:

$$\begin{aligned} \int_0^1 \exp(u^2) du &= \int_0^1 \exp(u^2) f(u) du \quad \text{where } f(u) = \begin{cases} 1 & 0 \leq u \leq 1 \\ 0 & \text{else} \end{cases} \\ &= \mathbb{E}_{f(u)}[\exp(u^2)]. \end{aligned}$$

That is: $\mathbb{E}[\exp(U^2)]$ where $U \sim U[0, 1]$.

Hence:

```
u=runif(5000)
mean(exp(u^2))
[1] 1.466655
```

Monte Carlo Integration

Now consider

$$\int_a^b g(x) dx$$

for finite a and $b > a$.

Create a transformation to map integration limits to $[0, 1]$:

Here $u = (x - a)/(b - a)$ and so $x = a + u(b - a)$. Hence

$$\int_a^b g(x) dx = \int_0^1 (b - a)g(a + u(b - a)) du$$

which is in the same form as before.

Note:

- ▶ Could alternatively simulate $x^{(1)}, \dots, x^{(N)} \sim U(a, b)$ and then compute $(b - a) \frac{1}{N} \sum_{i=1}^N g(x^{(i)})$.
- ▶ However, given samples from $U(0, 1)$ are transformed to produce $U(a, b)$, this is identical to the above.

Monte Carlo Integration

Example: Calculate

$$\int_0^2 \exp(x^2) dx.$$

Solution:

Make the transformation $u = x/2$:

$$\int_0^2 \exp(x^2) dx = 2 \int_0^1 \exp(4u^2) du.$$

Hence:

```
> u=runif(5000,0,2)
> mean(2*exp(u^2))
[1] 16.65154
```

```
> u=runif(5000)
> mean(2*exp(4*u^2))
[1] 17.01787
```

Monte Carlo Integration

Now suppose the range of integration is **infinite**:

$$\int_0^{\infty} g(x) dx.$$

Make the transformation $u = 1/(x + 1)$ so that $x = 1/u - 1$. Hence

$$\begin{aligned}\int_0^{\infty} g(x) dx &= \int_1^0 -\frac{1}{u^2} g\left(\frac{1}{u} - 1\right) du \\ &= \int_0^1 \frac{1}{u^2} g\left(\frac{1}{u} - 1\right) du\end{aligned}$$

(as $\int_0^1 f(x) dx = F(1) - F(0) = -(F(0) - F(1)) = -\int_1^0 f(x) dx$.)

Can now estimate as before.

Monte Carlo Integration

Example: Calculate

$$\int_0^{\infty} \exp(-x^2) dx.$$

Solution:

Make the transformation $u = 1/(x + 1)$:

$$\int_0^{\infty} \exp(-x^2) dx = \int_0^1 \frac{1}{u^2} \exp \left[- \left(\frac{1}{u} - 1 \right)^2 \right] du.$$

Hence:

```
> u=runif(5000)
> z=1/u-1
> mean(1/u^2*exp(-z^2))
[1] 0.8893568
```

Note: Exact answer = $\sqrt{\pi}/2 = 0.8862269$.

Monte Carlo Integration

Monte Carlo methods are most useful for approximating multidimensional integrals. E.g.

$$\int_0^1 \int_0^1 g(u_1, u_2) du_1 du_2.$$

Let $U_1, U_2 \sim U(0, 1)$. Their joint density is

$$f(u_1, u_2) = \begin{cases} 1 & \text{if } u_1 \in [0, 1], u_2 \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

And so

$$\begin{aligned} \mathbb{E}_f[g(U_1, U_2)] &= \int_0^1 \int_0^1 g(u_1, u_2) f(u_1, u_2) du_1 du_2 \\ &= \int_0^1 \int_0^1 g(u_1, u_2) du_1 du_2. \end{aligned}$$

Monte Carlo Integration

Hence, given $(u_1^{(1)}, u_2^{(1)}), \dots, (u_1^{(N)}, u_2^{(N)})$, with $u_1^{(i)}, u_2^{(i)} \sim U(0, 1)$, we have

$$\int_0^1 \int_0^1 g(u_1, u_2) du_1 du_2 \approx \frac{1}{N} \sum_{i=1}^N g(u_1^{(i)}, u_2^{(i)}),$$

with the approximation being exact as $N \rightarrow \infty$ under weak conditions.

Final comment

- ▶ This approach is remarkably easy to use, even in high dimensions.
- ▶ The price for this simplicity is that the variance is high.
- ▶ Techniques are available to improve the precision of this.
e.g. Importance Sampling.
- ▶ We look at importance sampling next week.

Outline

1. Mixture priors, prior predictive checking.
2. Multivariate models
3. Monte Carlo methods
 - Monte Carlo integration, rejection sampling.

Rejection Sampling

We want to sample from a distribution $f(x)$, but this is difficult to do.

Rejection sampling approach:

- ▶ Sample from a distribution, $x^* \sim g(x)$, from which it is easy to simulate
- ▶ Accept x^* with some probability, p .

If we choose p correctly, we can make accepted samples come from $f(x)$.

Example:

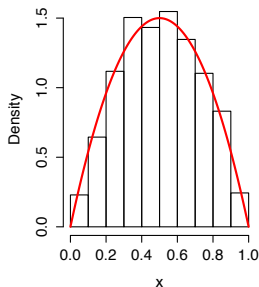
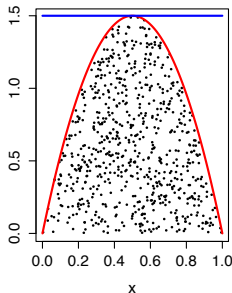
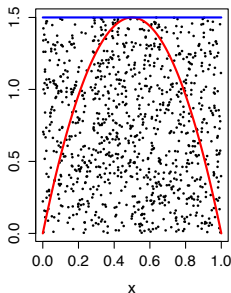
Obtain N samples from $\text{Beta}(2, 2)$.

Solution:

Inversion sampling is inefficient as requires numerical inversion of $F(x)$.

Instead, try rejection sampling.

Rejection Sampling



red = $f(x)$, blue = $Kg(x)$ Here: $g(x) = U(0, 1)$

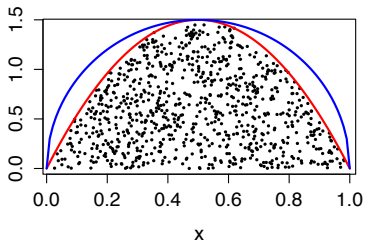
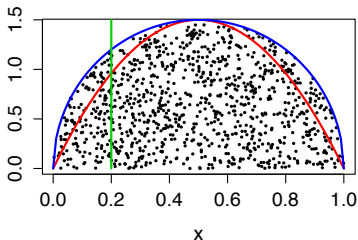
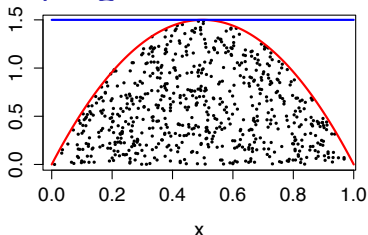
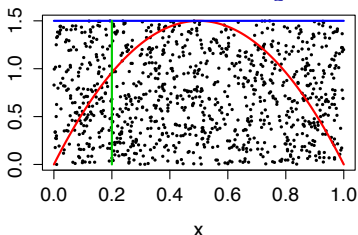
- ▶ Bound the density $f(x)$ by a bounding density ($Kg(x)$)
- ▶ Simulate points (x_i, y_i) uniformly under the bounding density:
i.e. $x_i \sim g(x)$ then $y_i \sim U(0, Kg(x_i))$
- ▶ Reject those points which do not lie under $f(x)$
i.e. accept x_i if $y_u \leq f(x_i)$
- ▶ x_i co-ordinates of accepted points are a sample from $f(x)$.

Rejection Sampling

Comments:

1. y_i plays role of accepting x_i with probability $f(x)/(Kg(x))$
(green line on next slide)
2. Efficiency of algorithm depends on number of rejections.
Bounding rectangle may be inefficient
3. To improve efficiency (and to allow for unbounded $f(x)$) modify bounding function to take any form $Kg(x)$, where
 - $g(x)$ is any density from which it is easy to simulate
 - $K > 0$ is some constant to be chosen
4. Algorithm acceptance probability is $= \frac{f(x)}{Kg(x)}$ (see previous pic)

Rejection Sampling



red = $f(x)$, blue = $kg(x)$

Top: $g(x) = \text{Beta}(1,1)$, efficiency = $681/1000 = 0.681$

Bottom: $g(x) = \text{Beta}(1.5,1.5)$, efficiency = $848/1000 = 0.848$

Rejection Sampling

Algorithm version 1

1. Generate x^* from $g(x)$
2. Generate y^* from $U(0, Kg(x^*))$
3. Accept x^* if $y^* \leq f(x^*)$

Algorithm version 2

1. Generate x^* from $g(x)$
2. Accept x^* with probability $\frac{f(x^*)}{Kg(x^*)}$

These are (obviously) identical.

Proof: Under Alg. 1, noting that $y^* \sim U(0, Kg(x^*))$,

$$Pr(\text{accept } x^*) = Pr(y^* \leq f(x^*)) = \int_0^{f(x^*)} \frac{1}{Kg(x^*)} dz = \frac{f(x^*)}{Kg(x^*)}. \checkmark$$

How to choose K for maximum efficiency?

We **require** $f(x) \leq Kg(x)$ (compulsory) $\forall x$.

$$K = \max_x f(x)/g(x) \quad \Rightarrow \quad \text{max acceptance probability} = 1.$$

Rejection Sampling

Why does this work?

Let $X \sim g(x)$ such that $f(x) \leq Kg(x)$, $\forall x$ for some K .

$f(x)$ is not required to be normalised! (i.e. $\int f(y)dy = c < \infty$)

Let $h(x) = f(x)/Kg(x)$ be the probability that x is accepted.

Then

$$\Pr(X \leq x \text{ and } X \text{ accepted}) = \int_{-\infty}^x h(y)g(y)dy$$

and

$$\Pr(X \text{ accepted}) = \int_{-\infty}^{\infty} h(y)g(y)dy$$

so

$$\begin{aligned}\Pr(X \leq x | X \text{ accepted}) &= \frac{\int_{-\infty}^x h(y)g(y)dy}{\int_{-\infty}^{\infty} h(y)g(y)dy} \\ &= \frac{\int_{-\infty}^x f(y)dy}{\int_{-\infty}^{\infty} f(y)dy}\end{aligned}$$

so that the accepted values have pdf f .

Rejection Sampling

Comments:

- Overall acceptance rate:

$$\Pr(X \text{ accepted}) = \int h(y)g(y)dy = \frac{1}{K} \int f(y)dy \propto \frac{1}{K}.$$

Hence want K as small as possible subject to $f(x) \leq Kg(x)$.
i.e. $K = \max_x f(x)/g(x)$

- The normalising constant in the denominator means that f needs only be known up to proportionality in order for algorithm to work.
- Algorithm efficiency depends on the difference between f and the bounding envelope Kg . (Recall Beta(2,2) example).
 $\Rightarrow$ If $g(x)$ gets closer to $f(x)$ then K can be smaller via $K = \max_x f(x)/g(x)$
Smaller K = greater efficiency.

Rejection Sampling

Example:

Simulate from the density

$$f(x) \propto \begin{cases} x(1-x)^3 & 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Use rejection sampling with

$$g(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(that is, use the uniform density on $[0, 1]$).

Rejection Sampling

To find the optimal K we want

$$K = \max_x \frac{f(x)}{g(x)} = \max_x f(x).$$

The derivative of $f(x)$ is

$$\begin{aligned} f'(x) &= x(-3(1-x)^2) + (1-x)^3 \\ &= (1-x)^2(1-4x) \end{aligned}$$

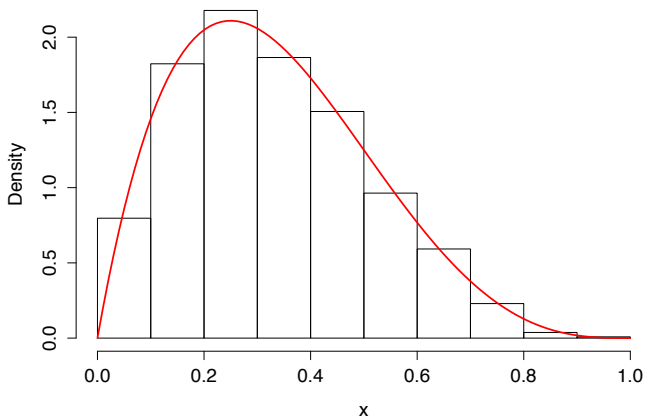
Hence $f'(x) = 0$ when $x = 1$ or $x = 1/4$.

Now $f(0) = f(1) = 0$ (local minimum) and
 $f(1/4) = (1/4) \cdot (3/4)^3 = 27/256 = K$ (local max as $f''(x) < 0$).

Hence

$$K = \max_x \frac{f(x)}{g(x)} = \frac{27}{256}.$$

Rejection Sampling



```
L=5000
K=27/256
xstar=runif(L)
ind=(runif(L)<(xstar*(1-xstar)^3/K))
hist(xstar[ind],probability=T,xlab="x",ylab="Density",main="")
xx=seq(0,1,length=100)
lines(xx,20*xx*(1-xx)^3,lwd=2,col=2)
```